

Automatic Fan Speed Controlling System

Project Report: Automatic Fan Speed Control System

1. Abstract

This project presents an automated solution for ceiling or table fan regulation to optimize power consumption and user comfort. By utilizing an **Arduino Uno** and **HC-SR04 Ultrasonic Sensor**, the system detects both the presence of individuals and their relative distance from the unit. The **L298N Motor Driver** modulates the DC motor speed using Pulse Width Modulation (PWM). The fan remains idle during zero occupancy and scales its RPM dynamically based on the number of people and their proximity, ensuring an efficient cooling experience.

2. Component Description

The system is built on a modular architecture where the Arduino acts as the central controller, processing sensor inputs to drive the mechanical output.

Component	Function
Arduino Uno	The primary microcontroller that executes the logic and controls the peripherals.
Ultrasonic Sensor (HC-SR04)	Uses sound waves to detect the number of people and their distance from the fan.
L298N Motor Driver	Interacts between the Arduino and the motor to manage high-current speed control.
DC Motor & Fan	The output assembly responsible for providing airflow.
Power Supply	Provides the external voltage required to drive the motor via the L298N module.

3. System Working & Methodology

The system operates based on a real-time feedback loop involving occupancy and proximity detection.

A. Occupancy-Based Activation

The ultrasonic sensor monitors the room for movement and presence. The system is programmed to distinguish the number of individuals within the sensor's field of view.

- **Idle State:** If the sensor detects no occupants, the Arduino sends a signal to the motor driver to keep the fan **OFF**, conserving energy.
- **Active State:** Upon detecting one or more persons, the system activates and begins calculating the required speed.

B. Dynamic Speed Regulation

The speed is not static; it adjusts based on two primary environmental factors:

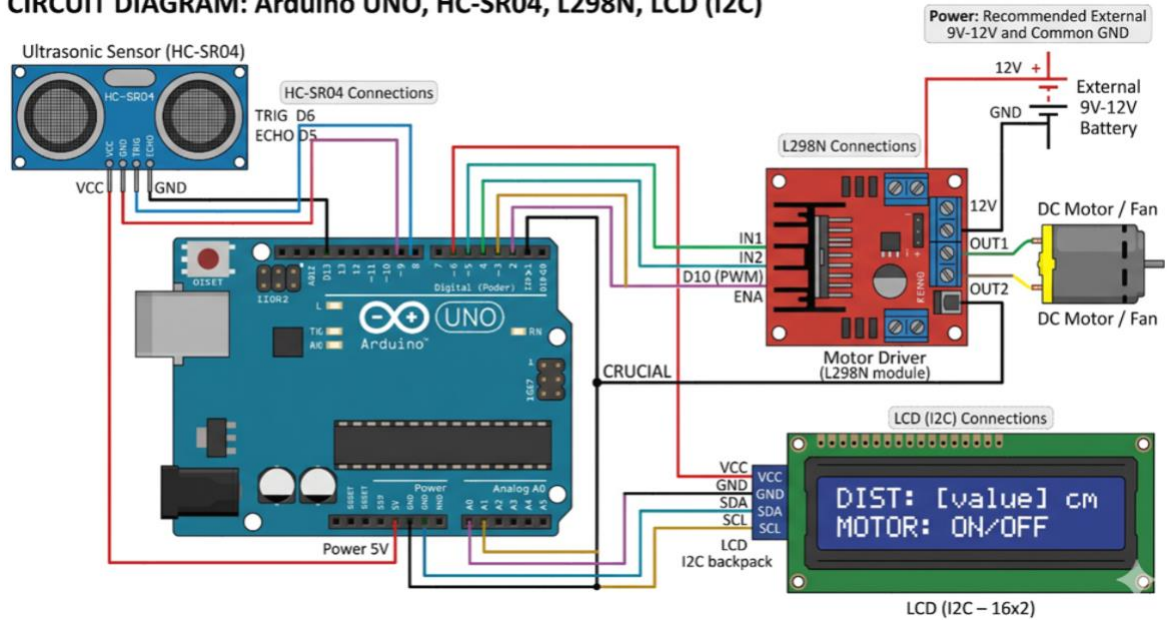
1. **Occupancy Density:** As the number of people in the room increases, the Arduino increases the fan speed to ensure sufficient air circulation for the entire group.
2. **Proximity Adjustment:** The system calculates the distance between the occupant and the fan. To ensure the airflow reaches the user effectively, the fan speed **increases** if the person is sitting further away and **decreases** as they move closer to the unit.

4. Advantages

- **Energy Efficiency:** Prevents the fan from running in empty rooms, significantly reducing electricity waste.
- **Autonomous Operation:** Eliminates the need for manual regulators or remote controls.
- **Enhanced Comfort:** Provides a personalized cooling experience by adjusting airflow based on exactly where the user is located.

5. Circuit Diagram

CIRCUIT DIAGRAM: Arduino UNO, HC-SR04, L298N, LCD (I2C)



6. Conclusion

The "Automatic Fan Speed Controlling System" successfully demonstrates how basic robotics components like the L298N and Arduino can be used to create smart home solutions. By prioritizing both user comfort and energy conservation, this project serves as a functional prototype for modern, responsive household appliances.